Shamima Hossain

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Research Interests

Post-training of language models, continually improving agents, human-ai collaboration and multilingual models.

Education

BRAC University
MSc in Computer Science — Fall 2025 – expected Feb 2027

- Thesis (in progress): exploring RL algorithms, and the conditions under which they succeed, that allow agents in automated scientific discovery to reuse previously acquired knowledge for efficient goal completion.

BRAC University
BSc in Computer Science — Jan 2017 – May 2021

- Thesis: *Smart Companion Agent for Mental Well-being Using Deep Learning and NLP* — transformer-based (BioBERT) detection of mental distress from physiological signals with a conversational monitoring interface. [thesis]

Summer & Winter Schools

- **ELLIS Summer School: AI for Research**, ELLIS Institute Finland, Aalto University, Aug 2026.
  - Selected as one of 100 participants for a program on AI-driven scientific discovery, trustworthy scientific agents, and uncertainty quantification.
  - Hackathon (LUMI supercomputer): **top 10** with a team project on auditing data quality in human-subject cognitive-science data using LLM agents.
- **ELLIS Winter School on Foundation Models (FoMo)**, ELLIS Unit Amsterdam, Mar 2026.
  - One of 260 early-career researchers from 128 institutions; lectures on LLM hallucination, multilingual models and cultural values, reasoning-model safety, and multimodal explainability.

Publications & Preprints

Author role is stated for each entry.

[1] **Shamima Hossain** (first author), Laila Sumiya Khan Nilo, Abdullah Al Mahmud . *Exploring DPO's Implicit Reward.* *Manuscript, 2026*
Tests DPO's "secretly a reward model" claim under a budget-matched setup (Qwen2.5-0.5B, one shared SFT checkpoint, five seeds): an explicit Bradley–Terry reward model wins in distribution, the gap does not widen under shift, and an untrained SFT log-probability baseline beats both trained scorers on RewardBench-Reasoning.
[PDF] [code] [checkpoints]

[2] **Shamima Hossain** (sole author). *Probing Structural Bias in Stable Diffusion 3: A Stereotype Content Model Analysis.* *Manuscript, 2026*
Probes Stable Diffusion3 against social psychology's warmth–competence model with a Bayesian errors-in-variables probabilistic graphical model over 6,600 generated portraits of 66 groups: SD3 encodes the structure reliably but collapses the "envy" quadrant (CEOs, lawyers), reframing text-to-image bias as stereotype geometry rather than frequency. Corpus and probing code released.
[PDF] [code] [corpus]

[3] **Shamima Hossain** (sole author). *Beyond Generation: Multi-Hop Reasoning for Factual Accuracy in Vision–Language Models.* *ICML 2025 Workshop (NewInML), poster*
[arXiv:2511.20531]

Research & Industry Experience

bKash Limited
Senior Research Engineer — Jun 2023 – present

- Owned data curation and the training pipeline for AVA, bKash's first LLM-based conversational agent, serving a national-scale fintech user base: designed the filtering and curation strategy end to end, and studied how curation choices propagated into model behaviour.

- Designed a task-grounded LLM dialogue system for financial transactions, studying how grounding and dialogue structure affect interaction reliability.
- Built distributed training pipelines for language and vision models on AWS SageMaker.
- Developed and maintain a production face anti-spoofing engine, with a focus on robustness under real-time constraints.

Selected Research Projects

Identifying Retrieved Content in RAG Outputs [code]

- Combined chain-of-thought prompting with token-level manipulation so that users can visually distinguish retrieved external knowledge from parametric knowledge in LLM outputs; showed that models can self-identify gaps in what they know.

Knowledge-Graph-Augmented Factual Image Captioning [code]

- Precursor to [3]: controlled comparison of hierarchical vs. unstructured knowledge injection (Qwen2-VL-2B-Instruct + GraphX), showing hierarchical structure measurably improves fact-checking of visual content.

Reusable Semantic Filters from Diffusion-Model Text Embeddings [code]

- Computed prompt-difference matrices in text-embedding space and showed they map to transferable semantic shifts across images.

Academic Service

- **Reviewer**, GlobalSouthInAI Workshop, NeurIPS 2026.
- **Artifact Evaluation Committee**, MLSys 2026.

Talks

- Paper reading: *Physics of Language Models, Part 3.3: Knowledge Capacity Scaling Laws* (Allen-Zhu & Li). Online. [recording]
- Paper reading: *Attention Is All You Need* (Vaswani et al.). Cohere Labs NLP series, online.

Open-Source Contributions

- **Hugging Face** (2022). Documentation contribution to `transformers` (zero-shot image classification); DreamBooth fine-tuning of Stable Diffusion for the Keras DreamBooth community event.

Technical Skills

ML	PyTorch, Hugging Face `transformers/datasets/TRL/diffusers`, open-instruct, PyMC
Infra	AWS SageMaker & Lambda, Google Cloud, distributed training in Ray
Programming Languages	Python, C;